In recent years, deep learning and computer vision models have significantly advanced the detection of diseases and pests in rice crops, with a strong focus on computational efficiency, offline operation, and compatibility with entry-level mobile devices. These advancements are critical for enhancing agricultural productivity and ensuring global food security by providing farmers with timely and accurate diagnostic tools.

## Benchmarking Executive Report

### Disease Detection Models

The accurate diagnosis of rice diseases is crucial for preventing significant crop yield losses. Traditional methods relying on manual inspection are often subjective, inefficient, and prone to errors. Deep learning models offer a more scientific and objective approach to disease identification.

### MobileNetV2

- **Description**: MobileNetV2 is a lightweight Convolutional Neural Network (CNN) architecture designed for mobile and embedded devices. It utilizes depthwise separable convolutions and an inverted residual structure with linear bottlenecks to reduce computational costs while maintaining high accuracy. The model aims to achieve efficiency by using fewer parameters, data, and space.
- **Advantages**:
    - **Computational Efficiency**: MobileNetV2 significantly reduces memory usage by 74%, floating-point operations (FLOPs) by 49%, and parameters by 50% compared to its original model, making it suitable for resource-constrained devices. It improves inference speed by 8.5% over the original MobileNetV2 model.
    - **High Accuracy**: MobileNetV2 models have achieved high accuracy in rice disease detection, with reported accuracies of 90.84%, 95%, 95.2% (validation accuracy), 97%, and 98% for various rice diseases. An improved MobileNetV2 model achieved 99.53% accuracy on the PlantVillage dataset with 0.9M parameters.
    - **Offline Operation and Mobile Compatibility**: Its lightweight nature and optimized architecture make it ideal for deployment on mobile phones, laptops, and other mobile terminals for real-time, on-field diagnosis without continuous internet connectivity. The model can be used in smartphone applications for convenient and efficient field diagnosis of rice diseases.

- **Limitations**:
  - MobileNetV2 may produce unsatisfactory results due to limitations such as high computational expenses and overfitting if not carefully designed. Early detection of rice diseases using MobileNetV2 still heavily relies on farmers' experience and visual assessment.
  - While effective for rice disease detection, its accuracy can be further improved, and it may require data augmentation to address dataset size limitations.

- **Technical Justification**: MobileNetV2's effectiveness stems from its inverted residual structure, which uses shortcut connections between thin bottleneck layers, and lightweight depthwise convolutions for filtering features to introduce non-linearity. Removing non-linearities in narrow layers helps maintain representational power. This design balances accuracy and efficiency, making it highly suitable for mobile deployment.

**EfficientNet**

- **Description**: EfficientNet is a deep learning model known for its scalability and efficiency, which allows it to achieve high accuracy with fewer parameters and FLOPs compared to other CNNs. It systematically scales network depth, width, and resolution using a compound scaling method.
- **Advantages**:
  - **High Accuracy**: EfficientNet models have demonstrated over 99% accuracy in plant leaf disease detection. EfficientNet-B0 has outperformed other models in rice grain quality control.
  - **Computational Efficiency**: EfficientNet models are designed to be efficient, making them suitable for resource-constrained devices like the NVIDIA Jetson Nano.
  - **Edge Device Compatibility**: EfficientNet models can be effectively deployed on edge computing devices such as NVIDIA Jetson Nano, facilitating real-time, on-site disease detection.

- **Limitations**:
  - While generally efficient, EfficientNet models can be computationally intensive, requiring high-performance GPUs for training and deployment. The performance of EfficientNet models can be influenced by factors such as the size and quality of the dataset, and the model's ability to generalize

to real-world scenarios with varying lighting and backgrounds can be challenging.
    - EfficientNet-B2 was the highest model of the EfficientNet family that could be trained on the Jetson Nano, indicating potential limitations for larger variants on smaller edge devices.

- **Technical Justification**: EfficientNet's strength lies in its compound scaling approach, which uniformly scales all dimensions of depth, width, and resolution using a fixed set of scaling coefficients. This method allows for better model performance and efficiency across different scales, optimizing for resource-limited environments.

**SSD-YOLO (Improved YOLOv8)**

- **Description**: SSD-YOLO is an improved YOLOv8-based model specifically designed for rice leaf disease detection. It enhances diagnostic accuracy and efficiency by incorporating a Squeeze-and-Excitation Network (SENet) attention mechanism, a Dynamic Sample (DySample) lightweight dynamic upsampling module, and Shape-aware Intersection over Union (ShapeIoU) Loss.
- **Advantages**:
    - **High Accuracy and Speed**: SSD-YOLO achieves average detection accuracies of 87.52% for rice brown spot, 99.48% for rice blast, and 98.99% for bacterial blight, improving upon original YOLOv8. It maintains a compact model size of 6MB and offers significant enhancements in detection accuracy and speed. The mAP and accuracy reached 95.32% and 95.19% respectively.
    - **Computational Efficiency**: The model is compact, reducing parameters and computational costs. The DySample module, for example, reduces model complexity and computational costs.
    - **Enhanced Feature Extraction**: SENetV2 optimizes the Bottleneck structure of YOLOv8, improving feature extraction capabilities and enhancing the model’s ability to recognize disease characteristics in complex backgrounds.
    - **Improved Localization**: ShapeIoU Loss enhances the model’s performance in complex environments by focusing on the shape and size characteristics of bounding boxes.

- **Limitations**:
    - Generalization to highly variable field conditions, such as lighting variations and occlusions, remains a challenge. The model's dependence

on high-quality training data can limit its applicability in resource-constrained regions where datasets are sparse.

- **Technical Justification**: SSD-YOLO integrates SENetV2 to recalibrate channel-wise feature responses, improving feature representation. DySample provides content-aware upsampling, accurately restoring detailed features and enhancing perception. ShapeIoU Loss, by considering bounding box shape and size, generates higher-quality bounding boxes and mitigates biases in evaluation.

**YOLO-CRD (Improved YOLOv5s)**

- **Description**: YOLO-CRD is an enhanced YOLOv5s network architecture for rice disease detection that incorporates a Convolutional Channel Attention Module, a Spatial Pyramid Pooling Cross-Stage Partial Channel module, and a Ghost module. These additions are designed to improve attention across image channels, enhance model generalization, and reduce model size.
- **Advantages**:
  - **High Accuracy**: Achieved a mean average precision of 90.2% and an accuracy of 90.4% on a test set. Specifically, it achieved 93.5% for bacterial brown spot and 94% for leaf rice fever.
  - **Computational Efficiency**: The Ghost module reduces the model size.
  - **Improved Generalization**: The Spatial Pyramid Pooling Cross-Stage Partial Channel module enhances the model's ability to generalize.
- **Limitations**: The search results do not explicitly provide information on the specific limitations of YOLO-CRD beyond general challenges in extracting early disease features in natural environments.
- **Technical Justification**: The Convolutional Channel Attention Module helps the model focus on important features, while the Spatial Pyramid Pooling Cross-Stage Partial Channel module allows for better handling of varying object scales. The Ghost module is used for lightweight feature extraction, reducing redundancy and computational costs.

**Pest Detection Models**

Timely and accurate detection of rice pests is vital for effective pest control, and for improving rice yield and quality. Manual identification is labor-intensive and prone to errors. Deep learning, particularly YOLO-based models, offers efficient and accurate solutions.

**Rice-YOLO (Improved YOLOv8)**

- **Description**: Rice-YOLO is an improved version of YOLOv8-N designed for fast and accurate rice pest detection. It integrates a lightweight detection head, deep supervision layers, and an improved dynamic upsampling module (DySample-hardtanh).
- **Advantages**:
  - **High Accuracy and Speed**: Achieved 78.1% mAP@0.5, 62.9% mAP@0.5:0.95, and 74.3% F1 scores on the IPR16 dataset. It offers the best detection accuracy and relatively fast inference speed compared to many traditional detection models like YOLOv3, YOLOv4, YOLOv5, YOLOv6, and YOLOv7.
  - **Computational Efficiency**: The lightweight detection head and dynamic upsampling module reduce parameters and computational load. It has 2.8 million parameters and 7.2 billion FLOPs.
  - **Enhanced Feature Perception**: The Diverse Branch Grouped (DBG) detection head improves the model's ability to perceive complex pest-related information. Deep supervision layers enhance gradient flow and convergence without adding inference burden.
- **Limitations**:
  - Despite improvements, challenges remain in detecting certain pest types due to high interclass similarity and significant appearance changes across different life stages. Crop pest datasets are scarce, and models may struggle with small interclass differences and complex backgrounds.
- **Technical Justification**: Rice-YOLO uses a Diverse Branch Grouped (DBG) detection head to enrich the feature space and reduce redundancy through group convolutions, improving inference without additional hardware overhead. Deep supervision with an auxiliary DBG head enhances training by providing additional gradients to intermediate layers, mitigating gradient vanishing. The DySample-hardtanh module improves sampling flexibility and feature upsampling by replacing the original dynamic range factor distribution function with the hardtanh function.

**Improved YOLOv8 with CBAM and BiFPN**

- **Description**: This model optimizes YOLOv8 by integrating the Convolutional Block Attention Module (CBAM) and the Bidirectional Feature Pyramid Network

(BiFPN) for feature fusion. CBAM enhances feature extraction, and BiFPN improves multi-scale feature fusion.

- **Advantages**:
  - **High Accuracy and Robustness**: Achieved mAP@0.5 and mAP@0.5:0.95 scores of 98.8% and 78.6%, respectively, representing increases of 2.8% and 2.35% over the original YOLOv8. It demonstrates greater robustness in handling complex backgrounds and varying lighting conditions.
  - **Enhanced Feature Extraction**: CBAM allows the model to focus more effectively on important features by adaptively weighting channel and spatial information.
  - **Improved Multi-scale Detection**: BiFPN enhances the integration of features from different scales, which is critical for detecting pests of various sizes.

- **Limitations**:
  - The search results do not explicitly provide specific limitations for this improved YOLOv8 model beyond the general challenges in pest detection, such as data scarcity and model complexity mentioned in the context of deep learning applications in agriculture.

- **Technical Justification**: CBAM integrates channel and spatial attention mechanisms to improve the model's ability to extract important features while suppressing irrelevant noise. BiFPN facilitates efficient and effective multi-scale feature fusion by combining features from different levels in a bi-directional pathway, leading to better detection of objects at various scales.

**RGC-YOLO (Improved YOLOv8n)**

- **Description**: RGC-YOLO is an efficient multi-scale rice pest detection model based on YOLOv8n. It integrates RepGhost for feature reuse, GhostConv for computational efficiency, and CBAM for enhanced feature extraction.
- **Advantages**:
  - **High Accuracy and Efficiency**: Achieves a precision of 86.2%, a recall of 90.8%, and an mAP50 of 93.2%. It reduces parameters by 33.2% and GFLOPs by 29.27% compared to the base YOLOv8n model.
  - **Fast Inference**: Deployed on an embedded Jetson Nano device, it reduces inference time per image by 21.3% compared to the base YOLOv8n model, reaching 170 milliseconds.

    - **Multi-scale Detection**: Developed for multi-scale rice pest and disease detection, addressing varying object sizes.

- **Limitations**: The search results do not explicitly provide limitations for RGC-YOLO beyond the general challenges in pest and disease detection in complex field environments.
- **Technical Justification**: RepGhost is a structural reparameterization module that enables implicit feature reuse, contributing to model efficiency. GhostConv layers replace standard convolutions to reduce computational cost and improve inference speed by generating more features from cheap operations. CBAM is incorporated into the backbone network to enhance the model's ability to extract important features by focusing on relevant channel and spatial information.

**GBiDC-PEST**

- **Description**: GBiDC-PEST is a lightweight, real-time multiclass tiny pest detection model optimized for mobile deployment. It is built upon the YOLO architecture and incorporates GhostNet for lightweight feature extraction, a Bi-directional Feature Pyramid Network (BiFPN) for enhanced multiscale feature fusion, Depthwise Convolution (DWConv) layers to reduce computational load, and a Convolutional Block Attention Module (CBAM) for precise feature focus.
- **Advantages**:
    - **High Accuracy and Speed**: Achieves a mean average precision (mAP) of 80.1% and an impressive processing speed of 161.3 frames per second (FPS).
    - **Lightweight and Mobile-friendly**: Significantly reduces model size to 2.8 MB, which is only 20% of the original model size, making it suitable for Android mobile deployment. It boasts a smaller size than YOLO series (v5–v10) and faster detection speed than v8s, v9c, v10m, and v10b.
    - **Robustness in Complex Environments**: Demonstrates enhanced performance in detecting pests against complex backgrounds.

- **Limitations**:
    - The model focuses specifically on four minuscule pests (wheat mites, sugarcane aphids, wheat aphids, and rice planthoppers). The generalization to a wider variety of agricultural pests might require fine-tuning and retraining on diversified datasets.
    - While effective, real-world agricultural fields present complex and unpredictable scenarios with varying light conditions and tiny pests

blending into backgrounds, which are still challenging for robust, high-accuracy detection.

- **Technical Justification**: GhostNet reduces model complexity by reconstructing the backbone network and stripping down redundant operations while preserving feature richness. BiFPN enhances multiscale feature fusion, which is crucial for detecting tiny objects. DWConv layers reduce parameters and operations, maintaining the lightweight nature of the model. CBAM improves the model's discriminative power by emphasizing informative spatial and channel-wise information, helping it focus on relevant pest features within cluttered backgrounds.

**MDP-YOLO (Improved YOLOv5s)**

- **Description**: MDP-YOLO is a lightweight rice pest detection model based on YOLOv5s, designed to address challenges like low detection accuracy, high computational complexity, and large model sizes in traditional methods. It uses ShuffleNetV2 as the backbone, introduces GhostConv, integrates a large-scale feature extraction layer, and incorporates CBAM.
- **Advantages**:
  - **High Accuracy and Efficiency**: Increases mean average precision by 5.5% while reducing FLOPs by 72.4%, parameters by 89.2%, and model size by 70.1% compared to the original YOLOv5s.
  - **Lightweight**: Achieves significant reductions in model size and computational cost, making it suitable for mobile deployment.
  - **Enhanced Feature Extraction**: Uses a large-scale feature extraction layer and CBAM to improve focus on regions of interest and detect small pest objects.

- **Limitations**: The search results do not explicitly provide specific limitations for MDP-YOLO beyond the general challenges in pest detection.
- **Technical Justification**: ShuffleNetV2 as the backbone network significantly reduces parameters and complexity. GhostConv replaces redundant convolutional layers, further decreasing computational complexity and size. The integration of a large-scale feature extraction layer enhances the algorithm's ability to detect small pest objects, and CBAM increases the focus on regions of interest.

**MobileNetv2-YOLOv4**

- **Description**: MobileNetv2-YOLOv4 is a lightweight model based on YOLOv4, which uses MobileNetv2 as the backbone network instead of CSPDarknet53 to minimize parameters and lighten the model. It also incorporates the Focal Loss function to balance positive and negative samples during training.
- **Advantages**:
    - **High Accuracy and Speed**: Achieves 90.50% mAP and 51.55 frames per second (FPS) on a 3-type field rice pest dataset. Compared to YOLOv4, the mAP is similar, but FPS is 206% higher, and model weights are reduced fivefold.
    - **Lightweight**: Significantly reduces the amount of parameters and model weights.
- **Limitations**: The search results do not explicitly provide specific limitations for MobileNetv2-YOLOv4 beyond addressing general issues of low pest detection effectiveness.
- **Technical Justification**: Replacing CSPDarknet53 with MobileNetv2 as the backbone network reduces parameters and lightens the model, leveraging MobileNetv2's efficiency. The Focal Loss loss function emphasizes the learning of difficult-to-classify samples, improving detection accuracy.

**Mobile-CANet**

- **Description**: Mobile-CANet is an effective real-time lightweight pest detection method based on convolutional neural networks. It focuses on enhancing feature extraction capabilities to address location information loss during feature extraction and incorporates transfer learning to improve generalization ability.
- **Advantages**:
    - **High Accuracy**: Achieves 85.3% accuracy on a publicly available dataset of rice pests.
    - **Lightweight**: Possesses only 3.312M parameters, making it suitable for deployment on edge devices with limited computing resources.
    - **Enhanced Feature Extraction and Generalization**: Develops a new network structure to enhance feature extraction capabilities and uses transfer learning to improve generalization.
- **Limitations**: The search results do not explicitly provide specific limitations for Mobile-CANet beyond the general challenges of deploying deep learning models on edge devices with limited computing resources.

- **Technical Justification**: The new network structure specifically designed for Mobile-CANet enhances feature extraction capabilities, crucial for accurate pest identification. Transfer learning is used to leverage knowledge from pre-trained models on large datasets, improving the model's ability to generalize to new pest images.

### Tiny-LiteNet

- **Description**: Tiny-LiteNet is a lightweight CNN optimized for edge applications, with built-in support for model explainability. It is part of a portable smart IoT device for real-time plant image acquisition and edge processing.
- **Advantages**:
  - **High Accuracy and Efficiency**: Achieves up to 98.6% accuracy, 98.4% F1-score, and 98.2% Recall. It has an 80 ms inference time and a compact model size of 1.2 MB with 1.48 million parameters.
  - **Edge Device Compatibility**: Outperforms traditional CNN architectures in efficiency and suitability for edge computing, with low power consumption and user-friendly design. It is integrated with Raspberry Pi 5 for edge processing.
- **Limitations**: The search results do not explicitly provide specific limitations for Tiny-LiteNet beyond addressing the challenges of maintaining crop health and the need for portable, cost-effective, and energy-efficient solutions for resource-constrained environments.
- **Technical Justification**: Tiny-LiteNet's design focuses on a lightweight architecture to achieve high accuracy with minimal computational resources, making it highly efficient for edge computing applications. Its optimization for low power consumption further enhances its practicality for field deployment on devices like Raspberry Pi.

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